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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Smart To-Do Task Manager with Priority Scheduling

Submitted by

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DECLARATION

I hereby declare that the project entitled "Smart To-Do Task Manager with Priority Scheduling" submitted in partial fulfillment of the requirements for the course **JAVA PROGRAMMING** is a **record of my original work** carried out under the guidance of **Mr.ARAVIND**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

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SRIKARAN K

CERTIFICATE

This is to certify that the project entitled **“Smart To-Do Task Manager with Priority Scheduling”** is a **bona fide work** carried out by SRIKARAN K **I ”B”/2025**, **BSC COMPUTER SCIENCE**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **JAVA PROGRAMMING**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: COIMBATORE

Date:10/04/2026

Submitted for the Viva-Voce Examination held on 10/04/2026

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TABLE OF CONTENTS

S. No.	Topics	Page. No.
1	Introduction	7
1.1	Objective	7
1.2	Overview	7
1.3	Features	8
2	**Program Analysis**	9
2.1	Problem Definition	9
2.2	Project Overview	9
2.3	Scopes and Limitations	10
3	**System Design**	11
3.1	Modules	11
4	**Methodology**	12
4.1	Problem Analysis	12

4.2	System Design	**12**
4.3	Program Development	**13**
4.4	Testing and Validation	**13**
4.5	Result and Documentation	**14**
4.6	Algorithm	**14**
4.7	Flowchart	**15**
5	**Source Code**	**17**
6	**Input and Output Screens**	**20**
7	**Discussion and Conclusion**	**22**
7.1	Discussion	**22**
7.2	Conclusion	**22**
8	**References**	**23**

ABSTRACT

The **Smart To-Do Task Manager with Priority Scheduling** is a Java-based application designed to improve task management and productivity. In today's busy environment, individuals often face difficulties in organizing their tasks and prioritizing important activities. Traditional to-do lists lack intelligent features to manage tasks effectively based on urgency and importance.

This project provides a smart solution by allowing users to create, update, delete, and view tasks in an organized manner. Each task is assigned a priority level such as high, medium, or low, and the system uses priority scheduling techniques to arrange tasks accordingly. This ensures that high-priority tasks are completed first, helping users manage their time efficiently and avoid missing deadlines.

The system is developed using core Java concepts such as object-oriented programming and data structures, ensuring efficiency and reliability. It is designed to be simple and user-friendly, making it suitable for a wide range of users.

In conclusion, this project offers an effective and practical approach to task management by combining simplicity with intelligent scheduling, thereby enhancing productivity and organization.

1.INTRODUCTION

In the modern digital era, effective task management has become an essential part of both personal and professional life. With increasing workloads, tight schedules, and multiple responsibilities, individu als often find it difficult to keep track of their tasks and complete them efficiently. Many people rely on traditional to-do lists or basic applications, but these systems lack intelligence in organizing and prioritizing tasks based on their importance and deadlines.

The **Smart To-Do Task Manager with Priority Scheduling** is a Java-based application developed to address these limitations. The main purpose of this system is to help users manage their daily tasks in a more structured and efficient manner by incorporating priority-based scheduling techniques. Unlike simple task lists, this application allows users to assign different priority levels—such as high, medium, and low—to each task. Based on these priorities, the system automatically arranges and schedules tasks so that more important tasks are handled first.

This project utilizes fundamental concepts of Java programming, including object-oriented principles like classes, objects, inheritance, and encapsulation. It also makes use of data structures such as lists or priority queues to store and organize tasks effectively. The scheduling mechanism ensures that tasks are processed in an order that maximizes productivity and minimizes delays. Additionally, the system provides features such as adding new tasks, updating existing tasks, deleting completed tasks, and viewing tasks in a sorted manner.

1.1 Objective

To develop a Java-based application for managing daily tasks efficiently

- To implement priority scheduling to organize tasks based on importance
- To allow users to add, update, delete, and view tasks easily
- To improve time management by completing high-priority tasks first
- To apply core Java concepts like object-oriented programming and data structures
- To create a simple, user-friendly system for better productivity

1.2 Overview

The **Smart To-Do Task Manager with Priority Scheduling** is a Java-based application developed to simplify and improve the way people manage their daily tasks. In today’s busy life, individuals often struggle to organize their work and complete important tasks on time. This

system provides an effective solution by offering a structured platform where users can easily add, update, delete, and view their tasks in one place.

A key feature of this application is its ability to assign priority levels such as high, medium, and low to each task. Based on these priorities, the system automatically arranges tasks so that the most important ones are handled first. This helps users focus on urgent activities, avoid missing deadlines, and make better use of their time. By organizing tasks in a systematic way, the application reduces confusion and improves overall productivity.

The project is built using core Java concepts, including object-oriented programming and data structures, making it both efficient and reliable. It is designed with a simple and user-friendly interface, allowing users of all levels to interact with it وهيق. Additionally, the system can be further enhanced with features like reminders and notifications, making it more powerful in the future.

Overall, the Smart To-Do Task Manager provides a smart, efficient, and practical approach to task management, helping users stay organized, reduce workload stress, and achieve their goals effectively.

1.3 Features

- **Add Tasks** – The system allows users to create new tasks by entering details such as task name, description, priority level, and deadline.
- **Update Tasks** – Users can modify existing tasks, including changing the task name, priority, or deadline as per their requirements.
- **Delete Tasks** – Unwanted or completed tasks can be removed from the system to keep the task list clean and organized.
- **View Tasks** – All tasks can be displayed in a structured format, making it easy for users to understand and manage their work.
- **Priority Assignment** – Each task can be assigned a priority level such as High, Medium, or Low based on its importance.
- **Priority Scheduling** – The system automatically arranges tasks according to their priority so that high-priority tasks are handled first.
- **Task Deadline Setting** – Users can assign due dates to tasks, helping them track deadlines and complete work on time.
- **Task Status Tracking** – Tasks can be marked as pending or completed, allowing users to monitor their progress easily.
- **User-Friendly Interface** – The application is designed with a simple and easy-to-use interface, suitable for all types of users.
- **Efficient Task Organization** – The system organizes tasks in a clear and systematic manner, reducing confusion and improving workflow.
- **Data Management** – Task details are stored and managed efficiently using appropriate data structures in Java.
- **Time Management Improvement** – By focusing on high-priority tasks, the system helps users save time and increase productivity.
- **Scalability** – The application can be extended with additional features like reminders

or notifications in the future.

2.PROGRAM ANALYSIS

The **Smart To-Do Task Manager with Priority Scheduling** manages tasks by storing and organizing them based on priority levels. It uses data structures like lists or priority queues to sort tasks so that high-priority tasks are handled first. The program supports operations like adding, updating, and deleting tasks. It is developed using object-oriented programming concepts, making the system efficient, simple, and easy to use.

2.1 Problem Definition

In today's fast-paced environment, managing multiple tasks efficiently has become a major challenge for individuals. Many people rely on traditional to-do lists or basic task management systems, which only allow users to store tasks without considering their importance or urgency. As a result, users often face difficulties in identifying which tasks should be completed first, leading to poor time management and missed deadlines.

Existing systems lack intelligent features such as priority-based scheduling, automatic task organization, and proper tracking of task status. This creates confusion when handling multiple tasks at the same time and reduces overall productivity. Users may spend more time deciding what to do next rather than actually completing tasks.

Therefore, there is a need for a smart system that can not only store tasks but also organize them based on priority. The **Smart To-Do Task Manager with Priority Scheduling** is designed to address these issues by providing an efficient way to manage tasks, prioritize important activities, and improve time management.

2.2 Project Overview

The **Smart To-Do Task Manager with Priority Scheduling** is a Java-based project developed to help users efficiently manage their daily tasks. The main objective of this system is to organize tasks based on their importance and ensure that high-priority tasks are completed first.

The application allows users to perform basic operations such as adding, updating, deleting, and viewing tasks. Each task can be assigned a priority level like high, medium, or low, and the system automatically arranges tasks using a priority scheduling technique. This ensures that important tasks are given more attention and are completed on time.

The project is implemented using core Java concepts such as object-oriented programming,

classes, and data structures like lists or priority queues. It is designed to be simple, user-friendly, and efficient, making it suitable for students and professionals.

Overall, this project provides a smart and structured approach to task management, helping users improve productivity, manage time effectively, and reduce workload stress.

2.3 Scopes and Limitations

Scope

The **Smart To-Do Task Manager with Priority Scheduling** has a wide scope in helping users manage their daily tasks efficiently. It can be used by students, professionals, or anyone who wants to organize their work in a structured manner. The system supports essential task management operations such as adding, updating, deleting, and viewing tasks. It improves time management by prioritizing tasks based on importance, ensuring that critical tasks are completed first. The project can also be extended in the future by adding advanced features like reminders, notifications, cloud storage, and mobile application support, making it more powerful and useful.

Limitations

Despite its usefulness, the system has some limitations. It is a basic application and may not support advanced features like real-time notifications or automatic reminders. The priority levels are fixed (high, medium, low), which may not cover all real-world scenarios. It may not support multi-user access or data synchronization across devices. Additionally, the system depends on user input for task details, so incorrect input can affect performance. The current version may also have a simple interface without advanced UI design.

3.SYSTEM DESIGN

The **Smart To-Do Task Manager with Priority Scheduling** is designed as a simple and efficient system that helps users manage tasks based on their priority. The system consists of three main components: the user interface, application logic, and data storage. The user interface allows users to interact with the system by adding, updating, deleting, and viewing tasks. The application logic processes the user input and applies priority scheduling to organize tasks. The data storage component stores task details such as task name, priority, deadline, and status.

The system is divided into modules including the task module, task management module, priority scheduling module, and user interface module. Each module performs a specific function, ensuring smooth operation of the system. Tasks are stored using data structures like lists or priority queues, which help in sorting tasks based on their priority levels.

The design follows object-oriented programming principles, where tasks are represented as objects with properties and methods. The system ensures that high-priority tasks are handled first, improving efficiency and time management. Overall, the design is simple, modular, and easy to maintain and extend.

3.1 Modules

1. **User Interface Module**
    - Shows menu (Add, View, Update, Delete)
    - Takes user input
    - Displays output

2. **Task Management Module**
    - Add new task
    - Update task
    - Delete task

3. **Priority Management Module**
    - Assign priority (High / Medium / Low)
    - Arrange tasks based on priority

4. **Data Storage Module**
    - Stores tasks in Array / ArrayList

5. **Sorting Module**
    - Sort tasks by priority
    - High → Medium → Low
6. **Exit Module**
    - Ends the program safely

4.METHODOLOGY

The **Smart To-Do Task Manager with Priority Scheduling** is developed using a systematic approach to ensure efficient task management. The methodology begins with collecting user requirements such as the need to add, update, delete, and prioritize tasks. Based on these requirements, the system is designed using object-oriented programming concepts in Java.

In the development phase, tasks are represented as objects with attributes like task name, priority level, deadline, and status. Suitable data structures such as ArrayList or Priority Queue are used to store and manage tasks. A priority scheduling algorithm is implemented to arrange tasks so that high-priority tasks are handled first, followed by medium and low-priority tasks.

The system then allows users to perform various operations through a simple interface. Whenever a task is added or modified, the system automatically updates and sorts the task list based on priority. Testing is carried out to ensure that all functionalities work correctly and efficiently.

4.1. Problem Analysis

Many users face difficulty in managing multiple tasks using traditional to-do lists, as they do not support priority-based organization. This leads to poor time management and missed important tasks. There is a need for a system that can organize tasks based on priority and help users complete important tasks first.Lack of proper **task tracking system** makes it hard to monitor completed and pending tasks Difficulty in **handling multiple tasks efficiently** leads to confusion and reduced productivity

4.2. System Design

The **Smart To-Do Task Manager with Priority Scheduling** is designed as a simple and efficient system that helps users manage tasks based on their priority. It consists of a user interface for

interaction, application logic to process tasks, and data storage to maintain task details. The system allows users to add, update, delete, and view tasks, while a priority scheduling mechanism automatically arranges tasks so that high-priority tasks are handled first. It uses object-oriented programming concepts and data structures like lists or priority queues to organize tasks effectively. The overall design is modular, user-friendly, and easy to maintain and extend.The system follows a **modular structure**, separating different functionalities for better performance It ensures **efficient task sorting** using priority-based scheduling techniques The design supports **real-time updates** when tasks are added or modified

4.3. Program Development

The development of the **Smart To-Do Task Manager with Priority Scheduling** is carried out using Java by following a step-by-step approach. Initially, the requirements of the system are identified, such as task creation, updating, deletion, and priority handling. Based on these requirements, the system is designed using object-oriented programming concepts.

In the development phase, a **Task class** is created to represent each task with attributes like task name, priority, deadline, and status. Then, a **Task Manager class** is implemented to handle operations such as adding, updating, deleting, and displaying tasks. Suitable data structures like **ArrayList or Priority Queue** are used to store and manage tasks efficiently.

Next, a **priority scheduling mechanism** is implemented to automatically sort tasks based on their priority levels (High, Medium, Low). The user interface is developed using simple console input/output to allow interaction with the system. The program is then tested to ensure all functionalities work correctly without errors.

4.4. Testing and Validation

Test Case ID	Description	Input	Expected Output

TC01	Add a new task	Task: "Study", Priority: High	Task is added successfully
TC02	View tasks	No input	Displays all tasks in priority order
TC03	Update task	Change "Study" to Medium priority	Task is updated successfully
TC04	Delete task	Delete "Study" task	Task is removed from list
TC05	Priority scheduling	Add tasks with High, Medium, Low	Tasks displayed as High → Medium → Low
TC06	Invalid input	Empty task name	Error message displayed
TC07	Mark task complete	Mark "Study" as completed	Task status updated to completed
TC08	Multiple tasks	Add 3–4 tasks	All tasks displayed correctly in sorted order

4.5. Result and Documentation

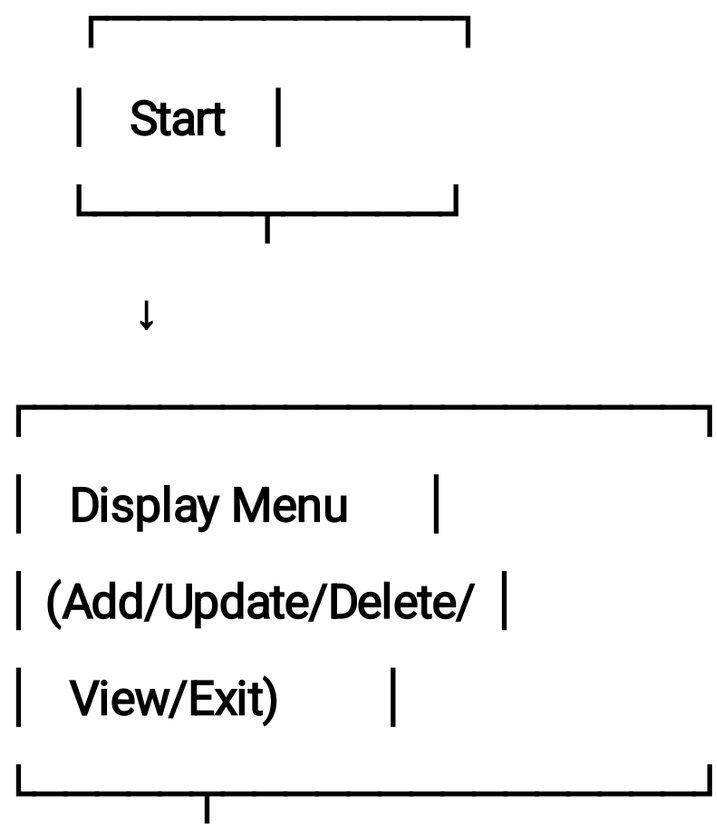
The **Smart To-Do Task Manager with Priority Scheduling** was successfully developed and tested, providing efficient task management with features like adding, updating, deleting, and prioritizing tasks. The system correctly organizes tasks based on priority, improving time management and productivity. The documentation includes all project details such as design, implementation, and usage, making the system easy to understand, use, and maintain.The system produces **accurate and reliable results** for all task operations It ensures **easy understanding through clear documentation and structure.**The project can be **easily enhanced with additional features in the future**

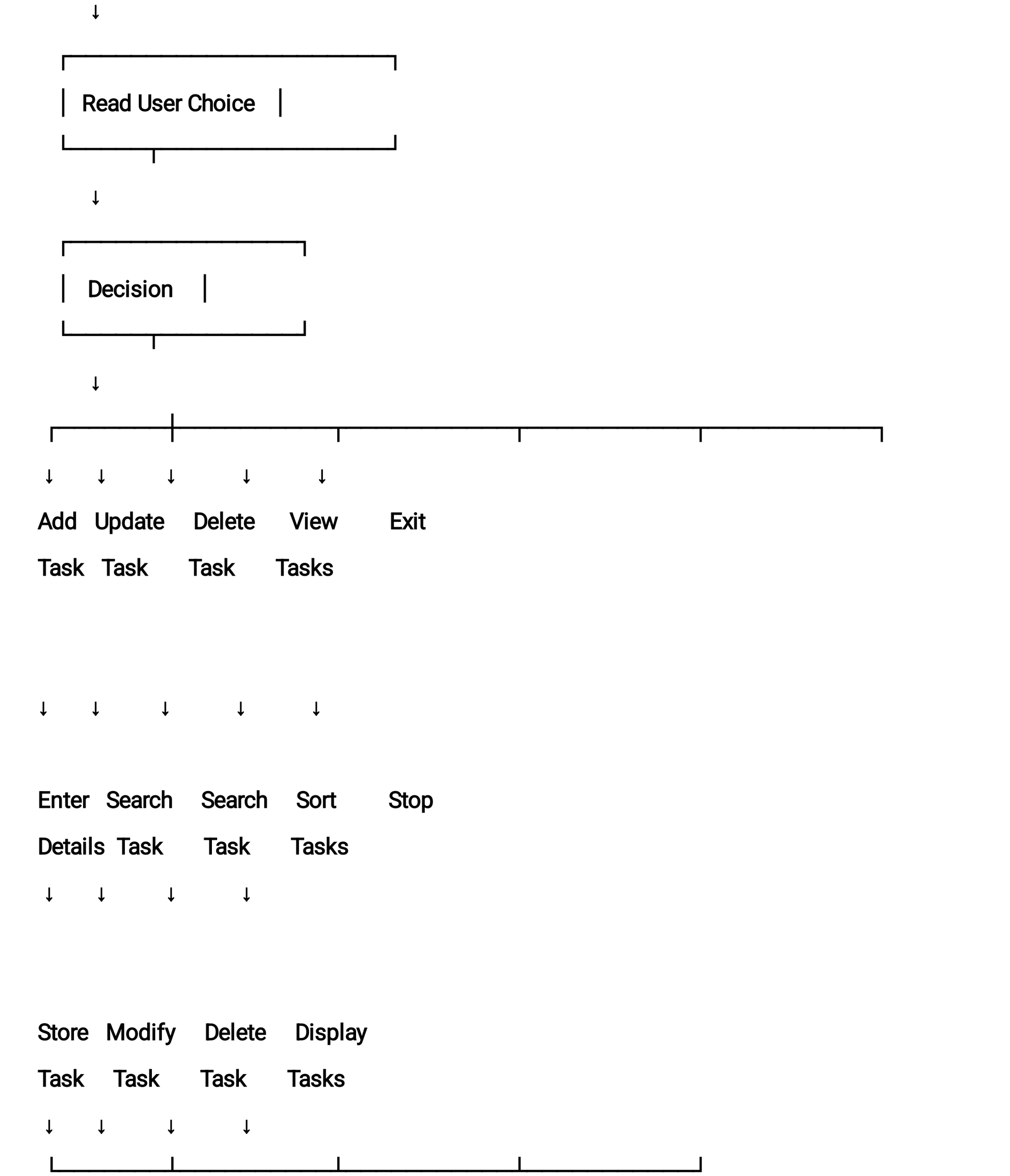
4.6. Algorithm

1. Start
2. Create empty task list
3. Display menu (Add / View / Update / Delete / Exit)

4. Read user choice
5. If **Add** → enter task & priority → store in list
6. If **View** → sort tasks by priority → display tasks
7. If **Update** →
    - enter task number
    - modify task name & priority
8. If **Delete** →
    - enter task number
    - remove task from list
9. If **Exit** → stop program
10. Repeat steps 3–9 until exit
11. End

4.7. Flowchart




Read User Choice
Decision
Add Task
Update Task
Delete Task
View Tasks
Exit
Enter Details
Search Task
Search Task
Sort Tasks
Stop
Store Task
Modify Task
Delete Task
Display Tasks
Back to Menu

5.SOURCE CODE

```
import java.util.*;

class Task implements Comparable<Task> {
    String name;
    String priority;
    String deadline;
    String status;
public Task(String name, String priority, String deadline) {
        this.name = name;
        this.priority = priority;
        this.deadline = deadline;
        this.status = "Pending";
    }

    public int compareTo(Task t) {
        return getPriorityValue(t.priority) - getPriorityValue(this.priority);
    }

    private int getPriorityValue(String p) {
        if (p.equalsIgnoreCase("High")) return 3;
        if (p.equalsIgnoreCase("Medium")) return 2;
        return 1;
    }

    public String toString() {
```

```
        return "Task: " + name + " | Priority: " + priority +
                " | Deadline: " + deadline + " | Status: " + status;
    }
}

public class SmartToDoManager {
    static PriorityQueue<Task> taskList = new PriorityQueue<>();
    static Scanner sc = new Scanner(System.in);

    public static void main(String[] args) {
        int choice;

        do {
            System.out.println("\n--- Smart To-Do Task Manager ---");
            System.out.println("1. Add Task");
            System.out.println("2. View Tasks");
            System.out.println("3. Delete Task");
            System.out.println("4. Mark Task Completed");
            System.out.println("5. Exit");
            System.out.print("Enter choice: ");
            choice = sc.nextInt();
            sc.nextLine();

            switch (choice) {
                case 1:
                    addTask();
                    break;
                case 2:
                    viewTasks();
                    break;
                case 3:
                    deleteTask();
                    break;
                case 4:
                    markCompleted();
                    break;
                case 5:
                    System.out.println("Exiting...");
                    break;
                default:
                    System.out.println("Invalid choice!");
            }

        } while (choice != 5);
    }
```

```
static void addTask() {
    System.out.print("Enter task name: ");
    String name = sc.nextLine();
    System.out.print("Enter priority (High/Medium/Low): ");
    String priority = sc.nextLine();
    System.out.print("Enter deadline: ");
    String deadline = sc.nextLine();

    taskList.add(new Task(name, priority, deadline));
    System.out.println("Task added successfully!");
}

static void viewTasks() {
    if (taskList.isEmpty()) {
        System.out.println("No tasks available.");
        return;
    }
    System.out.println("\nTasks (by priority):");
    PriorityQueue<Task> temp = new PriorityQueue<>(taskList);

    while (!temp.isEmpty()) {
        System.out.println(temp.poll());
    }
}

static void deleteTask() {
    System.out.print("Enter task name to delete: ");
    String name = sc.nextLine();

    Iterator<Task> it = taskList.iterator();
    boolean found = false;

    while (it.hasNext()) {
        Task t = it.next();
        if (t.name.equalsIgnoreCase(name)) {
            it.remove();
            found = true;
            System.out.println("Task deleted!");
            break;
        }
    }

    if (!found) {
        System.out.println("Task not found!");
```

```
    }
}

static void markCompleted() {
    System.out.print("Enter task name to mark completed: ");
    String name = sc.nextLine();

    for (Task t : taskList) {
        if (t.name.equalsIgnoreCase(name)) {
            t.status = "Completed";
            System.out.println("Task marked as completed!");
            return;
        }
    }
    System.out.println("Task not found!");
  }
}
```

6.INPUT AND OUTPUT SCREENS

Input & Output

Sample Run 1 – Add Task

Input:

1

Study Java

High

15-04-2026

Output:

— Smart To-Do Task Manager —

1. Add Task

2. View Tasks

3. Delete Task

4. Mark Task Completed

5. Exit

Enter choice: 1

Enter task name: Study Java

Enter priority (High/Medium/Low): High

Enter deadline: 15-04-2026

Task added successfully!

Sample Run 2 – View Tasks

Input:

2

Output:

Tasks (by priority):

Task: Study Java | Priority: High | Deadline: 15-04-2026 | Status: Pending

Sample Run 3 – Mark Task Completed

Input:

4

Study Java

Output:

Enter task name to mark completed: Study Java

Task marked as completed!

Sample Run 4 – Delete Task

Input:

3

Study Java

Output:

Enter task name to delete: Study Java

Task deleted!

Sample Run 5 – Exit

Input:

5

Output:

Exiting...

7.DISCUSSION & CONCLUSION

7.1 Discussion

The Smart To-Do Task Manager with a Priority System was developed to address the limitations of traditional task management methods by introducing structured organization and intelligent prioritization. The system allows users to create, update, delete, and manage tasks efficiently while assigning priority levels such as High, Medium, and Low.

The core functionality of the application is built using Java, ensuring platform independence and object-oriented design principles. The use of classes and data structures such as lists or priority queues enables efficient storage and retrieval of tasks based on their urgency and importance. Tasks with higher priority are displayed or executed before lower-priority ones, improving productivity and decision-making.

7.2 Conclusion

The Smart To-Do Task Manager with a Priority System successfully demonstrates how Java can be used to build a practical and efficient task management application. The project achieves its primary goal of organizing tasks and improving productivity through priority-based scheduling.

By implementing core concepts such as object-oriented programming, data structures, and user interaction, the system provides a strong foundation for real-world applications. It simplifies task handling and helps users manage their time more effectively.

In the future, the project can be enhanced by integrating features such as cloud storage, mobile support, notifications, and intelligent task suggestions. These improvements would make the system more scalable and user-friendly.

Overall, this project highlights the importance of structured task management and showcases how a simple yet powerful priority system can significantly improve efficiency and workflow.

8.REFERENCES

The following resources were used during the development of this project:

Herbert Schildt, *Java: The Complete Reference*, McGraw-Hill Education

- E. Balagurusamy, *Programming with Java*, McGraw-Hill
- Oracle Documentation – Java Programminghttps://docs.oracle.com/javase/tutorial/
- GeeksforGeeks – Java Programming and Data Structureshttps://www.geeksforgeeks.org/java/
- TutorialsPoint – Java Tutorial
- W3Schools – Java Tutorialhttps://www.w3schools.com/java/